

THE PUBLIC IS MORE FAMILIAR WITH BAD DESIGN THAN GOOD DESIGN. IT IS, IN EFFECT, CONDITIONED TO PREFER BAD DESIGN, BECAUSE THAT IS WHAT IT LIVES WITH. THE NEW BECOMES THREATENING, THE OLD REASSURING.

PAUL RAND, *DESIGN, FORM, AND CHAOS*

A DESIGNER KNOWS THAT HE HAS ACHIEVED PERFECTION NOT WHEN THERE IS NOTHING LEFT TO ADD, BUT WHEN THERE IS NOTHING LEFT TO TAKE AWAY.

ANTOINE DE SAINT-EXUPÉRY

...THE DESIGNER OF A NEW SYSTEM MUST NOT ONLY BE THE IMPLEMENTOR AND THE FIRST LARGE-SCALE USER; THE DESIGNER SHOULD ALSO WRITE THE FIRST USER MANUAL... IF I HAD NOT PARTICIPATED FULLY IN ALL THESE ACTIVITIES, LITERALLY HUNDREDS OF IMPROVEMENTS WOULD NEVER HAVE BEEN MADE, BECAUSE I WOULD NEVER HAVE THOUGHT OF THEM OR PERCEIVED WHY THEY WERE IMPORTANT.

DONALD E. KNUTH

THE TUFTE-LATEX DEVELOPERS

PUBLISHER OF THIS BOOK

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*Dedicated to those who appreciate $\text{\LaTeX}$
and the work of Edward R. Tufte and Donald E. Knuth.*

Introduction

This sample book discusses the design of Edward Tufte's books¹ and the use of the tufte-book and tufte-handout document classes.

¹ Edward R. Tufte. *The Visual Display of Quantitative Information*. Graphics Press, Cheshire, Connecticut, 2001. ISBN 0-9613921-4-2; Edward R. Tufte. *Envisioning Information*. Graphics Press, Cheshire, Connecticut, 1990. ISBN 0-9613921-1-8; Edward R. Tufte. *Visual Explanations*. Graphics Press, Cheshire, Connecticut, 1997. ISBN 0-9613921-2-6; and Edward R. Tufte. *Beautiful Evidence*. Graphics Press, LLC, first edition, May 2006. ISBN 0-9613921-7-7

A: Modules

Modules are a simultaneous generalization of vector spaces, ideals and quotient rings. The easiest way (for me) to remember the definition is that a module is a vector space except that the scalar multiplication comes from a ring.

Definition 1 (*R*-module). Let R be a ring and an abelian group M together with a map $*$: $R \times M \rightarrow M$ satisfying

$$r * (s * m) = (rs) * m \tag{1}$$

$$r * (m + n) = r * m + r * n \tag{2}$$

$$(r + s) * m = r * m + s * m \tag{3}$$

is an **left R -module**. If the multiplication is on the right then we call M a right R -module if R is commutative then we extend the commutativity to multiplication with M and just call M and R -module. Usually the map $*$ is denoted by juxtaposition because either it is identical to the multiplication in R or the distinction is made by adopting different symbols for elements of R and M (e.g. Av).

B: Review of Linear Algebra

The determinant

Definition 2 (Determinant). The determinant the unique alternating multilinear map

$$\det : (\mathbb{F}^n)^n \rightarrow \mathbb{F}$$

s.t.

$$\det(e_1, e_2, \dots, e_n) = 1.$$

It is also defined for matrices. For an $n \times n$ matrix A

$$\det(A) := \det\left(\sum_j A_{j1}e_1, \dots, \sum_j A_{jn}e_n\right)$$

Lemma 1. if vectors $v_1, \dots, v_n$ are linearly dependent then

$$\det(v_1, \dots, v_n) = 0$$

Proof. By alternation $\det(\dots, a, \dots, a, \dots) = 0$, that is if any two vectors among the arguments are equal then the determinant is equal to 0.

Hence if $v_1 = \sum_{i=2}^n \alpha_i v_i$ then $\det(v_1, v_2, \dots, v_n) = \sum_{i=2}^n \alpha_i \det(v_i, v_2, \dots, v_n) = 0$

□

Proposition (Leibnitz formula for determinant).

$$\det A := \sum_{\sigma \in S_n} \text{sign}(\sigma) \prod_{i=1}^n A_{ii^\sigma}$$

Definition 3 (Minors, Cofactors). Let A be an $n \times m$ matrix a minor of order p is the determinant

$$A \begin{pmatrix} i_1, \dots, i_p \\ j_1, \dots, j_p \end{pmatrix} := \det \begin{bmatrix} A_{i_1 j_1} & \dots & A_{i_1 j_p} \\ \vdots & & \vdots \\ A_{i_p j_1} & \dots & A_{i_p j_p} \end{bmatrix}$$

For a square matrix A of order n the ij -th principal minor is the order $n-1$ minor obtained by deleting from A the i -th row and j -th column. We denote it

$$A \begin{pmatrix} \hat{i} \\ \hat{j} \end{pmatrix} := A \begin{pmatrix} 1, \dots, i-1, i+1, \dots, n \\ 1, \dots, j-1, j+1, \dots, n \end{pmatrix}$$

The ij -th cofactor of A is then the quantity $C_{ij} := (-1)^{i+j} A(\hat{i}, \hat{j})$.

Proposition (Laplace's formula).

$$\forall i, j \quad \det(A) = \sum_{s=1}^n A_{is} C_{is} = \sum_{t=1}^n A_{tj} C_{tj}$$

Proof. By applying Laplace's formula recursively notice that once fully expanded each summand corresponds to a permutation of $[1, n]$. What remains to be determined is how the product of signs appearing in front of each summand relates to the sign of the corresponding permutation. To do so we turn now to a problem whose relation to our aim may not be immediately apparent.

consider a permutation $(m_1, m_2, \dots, m_n)$. Writing $m_{k_i} = i$, we can compute the sign of this permutation by first reordering it as $(1, m_2, \dots, m_{k_1-1}, m_{k_1+1}, \dots, m_n)$, by applying $k_1 - 1$ transpositions each moving 1 one closer to the first position. Now by the same process we transform the list to $(1, 2, m_{k_3}, \dots, m_{k_n})$. If $k_2 > k_1$ this is achieved by $k_2 - 2$ transpositions, while if $k_2 < k_1$ it is achieved by $k_2 - 2 + 1 = k_2 - 1$ transpositions (because two has shifted up by one in the list). And in general it takes $k_r - r + |\{i < r | k_i > k_r\}|$ transpositions to move r to the r -th position in our manner. And so the sign of our permutation is

$$\prod_{r=1}^n -1^{k_r - r + |\{i < r | k_i > k_r\}|}$$

□

Corollary 1. Let A be a $n \times n$ matrix and v a n vector. Then the quantities

$$\sum_{i=1}^n v_i C_{ij}, \quad \sum_{j=1}^n v_j C_{ij}$$

are determinants of matrices obtained by replacing the j -th column or i -th row of A by v respectively. Because the determinant of a matrix with a repeated row or column is 0 this tells us that in particular:

$$\sum_{s=1}^n A_{is} C_{js} = \sum_{s=1}^n A_{si} C_{sj} = \delta_{ij} \det(A)$$

Definition 4 (Adjugate matrix). The adjugate of a $n \times n$ matrix A is an $n \times n$ matrix with entries

$$\text{adj}(A)_{ij} = C_{ji}$$

It has the property (Corollary 1) that

$$\text{adj}(A)A = A \text{adj}(A) = \det(A)I$$

Hence if A is invertible $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$, while if A is singular then $\text{adj}(A)$ is a zero divisor of A .

C: Matrix polynomials and clasification of pencils

In this appendix we develop the algebra of matrices with coefficients in $k[\lambda]$ (k an algebraically closed field).

Matrix polynomials, polynomial matrices

By a matrix polynomial we mean a matrix whose entires are univariate polynomials i.e. an $A \in M(n, m, k[\lambda])$. By grouping together like powers of lambda A can also be regarded as a polynomial with matrix coefficients i.e. $A = A_0\lambda^m + A_1\lambda^{m-1} \dots A_{m-1}\lambda + A_m \in M(n, m, k)[\lambda]$. A square matrix polnmial is called **regular** if $|A_0| \neq 0$

The division algorithm

Let $A(\lambda) = \sum_{i=0}^r A_i \lambda^{r-i}$, $B(\lambda) = \sum_{i=0}^s B_i \lambda^{s-i}$ by $n \times n$ matrix polynomials and B regular. Dividing A by B means expressing A as

$$A = Q(\lambda)B(\lambda) + R(\lambda)$$

where Q is of degree $r - s$ and R is of degree less than s . The algorithm proceeds in the same way as division of scalar polynomials. The thing to notice is that $\lambda^{r-s} A_0 B_0^{-1} B = Q_0(\lambda)B$ is a degree s polynomial with leading term $A_0 \lambda^s$, so $R_0 = A - Q(\lambda)B$ produces a polynomial of degree less than r . If $\deg(R_0) > s$ we repeat this process with R_0 in the place of A to produce an R_1 and so on until we get an R_t of degree less than s .

Algorithm 1 Left quotient

```

1: procedure LEFT QUOTIENT( $A, B$ )  $\triangleright$  Returns  $Q, R$  s.t.  $A = QB + R$ 
2:    $Q = 0$ 
3:    $B_0^{-1} = \text{lc}(B)^{-1}$ 
4:   while  $\deg(A) \geq \deg(B)$  do
5:      $A \leftarrow A - \lambda^{\deg(A)-\deg(B)} \text{lc}(A) B_0^{-1} B$ 
6:      $Q \leftarrow Q + \lambda^{\deg(A)-\deg(B)} \text{lc}(A) B_0^{-1}$ 
7:   end while
8:   return  $(Q, A)$   $\triangleright$  By the end of the while loop  $A = R$ 
9: end procedure

```

Charactesitic Polynomial, Generalized Bezouts theorem and Cayley-Hamilton theorem

A matrix polynomial are all familiar with from linear algebra is $\lambda I - A$ where A is a scalar $n \times n$ matrix. It's determinant $\Delta(\lambda; A) = |\lambda I - A|$ is the characteristic polynomial of A . In this section we shall show that if a matrix polynomial $P(\lambda)$ has A as a root then we can factor $\lambda I - A$ from it on either the left or right.

Theorem 1 (Generalized Bezouts theorem / G.B.T). *Let $P(\lambda) = Q(\lambda)(\lambda I - A) + R(\lambda) = (\lambda I - A)\hat{Q}(\lambda) + \hat{R}(\lambda)$ be a matrix polynomial and $R(\lambda), \hat{R}(\lambda)$ it's left, right remainder upond divison by $\lambda I - A$. Then $R = P(A), \hat{R} = \hat{P}(A)$*

Proof. Set $P = P_0\lambda^m + P_1\lambda^{m-1} + \dots + P_m = R_0$ and compute the left quoutient:

$$\begin{aligned} R_1 &= R_0 - \lambda^{m-1}P_0(\lambda I - A) = (P_0A + P_1)\lambda^{m-1} + P_2\lambda^{m-2} + \dots + P_m \\ R_2 &= R_1 - \lambda^{m-2}(P_1 + P_0A)(\lambda I - A) = (P_0A^2 + P_1A + P_2)\lambda^{m-2} + P_3\lambda^{m-3} + \dots + P_m \\ &\dots \\ R_m &= R_{m-1} - (P_0A^{m-1} + P_1A^{m-2} + \dots + P_{m-1}A)(\lambda I - A) = (P_0A^m + P_1A^{m-1} + \dots + P_{m-1}A)\lambda^0 + P_m = P(A) \end{aligned}$$

A similar computation of the right quoutient show the other claim. $\square$

Corolarry 2 (Corolarry to G.B.T). $F(A) = \hat{F}(A) = 0 \iff F(\lambda)$ is divisible by $(\lambda I - A)$

Theorem 2 (Cayley-Hamilton). *An $n \times n$ scalar matrix satifies it's own characteristic equation that is:*

$$\Delta(A; A) = 0$$

Proof. The trick is to consider the adjugate of $\lambda I - A$; it yields us the equations

$$\text{adj}(\lambda I - A)(\lambda I - A) = (\lambda I - A)\text{adj}(\lambda I - A) = \Delta(\lambda; A)I$$

Showing that $\Delta(\lambda; A)I$ is divisible by A on both sides. By G.B.T we have

$$\Delta(A; A) = 0$$

$\square$

The Design of Tufte's Books

THE PAGES of a book are usually divided into three major sections: the front matter (also called preliminary matter or prelim), the main matter (the core text of the book), and the back matter (or end matter).

THE FRONT MATTER of a book refers to all of the material that comes before the main text. The following table shows a list of material that appears in the front matter of *The Visual Display of Quantitative Information*, *Envisioning Information*, *Visual Explanations*, and *Beautiful Evidence* along with its page number. Page numbers that appear in parentheses refer to folios that do not have a printed page number (but they are still counted in the page number sequence).

Page content	Books			
	VDQI	EI	VE	BE
Blank half title page	(1)	(1)	(1)	(1)
Frontispiece ²	(2)	(2)	(2)	(2)
Full title page	(3)	(3)	(3)	(3)
Copyright page	(4)	(4)	(4)	(4)
Contents	(5)	(5)	(5)	(5)
Dedication	(6)	(7)	(7)	7
Epigraph	–	–	(8)	–
Introduction	(7)	(9)	(9)	9

² The contents of this page vary from book to book. In *VDQI* this page is blank; in *EI* and *VE* this page holds a frontispiece; and in *BE* this page contains three epigraphs.

The design of the front matter in Tufte's books varies slightly from the traditional design of front matter. First, the pages in front matter are traditionally numbered with lowercase roman numerals (*e.g.*, i, ii, iii, iv, . . .). Second, the front matter page numbering sequence is usually separate from the main matter page numbering. That is, the page numbers restart at 1 when the main matter begins. In contrast, Tufte has enumerated his pages with arabic numerals that share the same page counting sequence as the main matter.

There are also some variations in design across Tufte's four books. The page opposite the full title page (labeled "frontispiece" in the above table) has different content in each of the books. In *The Visual Display of Quantitative Information*, this page is blank; in *Envisioning Information* and *Visual Explanations*, this page holds a frontispiece; and in *Beautiful Evidence*, this page contains three epigraphs.

The dedication appears on page 6 in *VDQI* (opposite the introduction), and is placed on its own spread in the other books. In *VE*, an epigraph shares the spread with the opening page of the introduction.

None of the page numbers (folios) of the front matter are expressed except in *BE*, where the folios start to appear on the dedication page.

THE FULL TITLE PAGE of each of the books varies slightly in design. In all the books, the author's name appears at the top of the page, the title is set just above the center line, and the publisher is printed along the bottom margin. Some of the differences are outlined in the following table.

Feature	<i>VDQI</i>	<i>EI</i>	<i>VE</i>	<i>BE</i>
Author				
Typeface	serif	serif	serif	sans serif
Style	italics	italics	italics	upright, caps
Size	24 pt	20 pt	20 pt	20 pt
Title				
Typeface	serif	serif	serif	sans serif
Style	upright	italics	upright	upright, caps
Size	36 pt	48 pt	48 pt	36 pt
Subtitle				
Typeface	—	—	serif	—
Style	—	—	upright	—
Size	—	—	20 pt	—
Edition				
Typeface	sans serif	—	—	—
Style	upright, caps	—	—	—
Size	14 pt	—	—	—
Publisher				
Typeface	serif	serif	serif	sans serif
Style	italics	italics	italics	upright, caps
Size	14 pt	14 pt	14 pt	14 pt

THE TABLES OF CONTENTS in Tufte's books give us our first glimpse of the structure of the main matter. *The Visual Display of Quantitative Information* is split into two parts, each containing some number of chapters. His other three books only contain chapters—they're not broken into parts.

Edward R. Tufte

The Visual Display
of Quantitative Information

SECOND EDITION

Graphics Press · Cheshire, Connecticut

Edward R. Tufte

Envisioning Information

Graphics Press · Cheshire, Connecticut

Edward R. Tufte

Visual Explanations

Images and Quantities, Evidence and Narrative

Graphics Press · Cheshire, Connecticut

EDWARD R. TUFTE

BEAUTIFUL EVIDENCE

GRAPHICS PRESS LLC

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Typefaces

Tufte’s books primarily use two typefaces: Bembo and Gill Sans. Bembo is used for the headings and body text, while Gill Sans is used for the title page and opening epigraphs in *Beautiful Evidence*. Since neither Bembo nor Gill Sans are available in default L^AT_EX installations, the Tufte-L^AT_EX document classes default to using Palatino and Helvetica, respectively. In addition, the Bera Mono typeface is used for monospaced type.

The following font sizes are defined by the Tufte-L^AT_EX classes:

L ^A T _E X size	Font size	Leading	Used for
\tiny	5	6	sidenote numbers
\scriptsize	7	8	–
\footnotesize	8	10	sidenotes, captions
\small	9	12	quote, quotation, and verse environments
\normalsize	10	14	body text
\large	11	15	B-heads
\Large	12	16	A-heads, TOC entries, author, date
\LARGE	14	18	handout title
\huge	20	30	chapter heads
\Huge	24	36	part titles

Table 1: A list of L^AT_EX font sizes as defined by the Tufte-L^AT_EX document classes.

Headings

Tufte’s books include the following heading levels: parts, chapters,³ sections, subsections, and paragraphs. Not defined by default are: sub-subsections and subparagraphs.

³ Parts and chapters are defined for the tufte-book class only.

Heading	Style	Size
Part	roman	24/36×40 pc
Chapter	italic	20/30×40 pc
Section	italic	12/16×26 pc
Subsection	italic	11/15×26 pc
Paragraph	italic	10/14

Table 2: Heading styles used in *Beautiful Evidence*.

Paragraph Paragraph headings (as shown here) are introduced by italicized text and separated from the main paragraph by a bit of space.

Environments

The following characteristics define the various environments:

Environment	Font size	Notes
Body text	10/14×26 pc	
Block quote	9/12×24 pc	Block indent (left and right) by 1 pc
Sidenotes	8/10×12 pc	Sidenote number is set inline, followed by word space
Captions	8/10×12 pc	

Table 3: Environment styles used in *Beautiful Evidence*.

On the Use of the *tufte-book* Document Class

The Tufte- $\text{\LaTeX}$ document classes define a style similar to the style Edward Tufte uses in his books and handouts. Tufte's style is known for its extensive use of sidenotes, tight integration of graphics with text, and well-set typography. This document aims to be at once a demonstration of the features of the Tufte- $\text{\LaTeX}$ document classes and a style guide to their use.

Page Layout

Headings

This style provides A- and B-heads (that is, `\section` and `\subsection`), demonstrated above.

If you need more than two levels of section headings, you'll have to define them yourself at the moment; there are no pre-defined styles for anything below a `\subsection`. As Bringhurst points out in *The Elements of Typographic Style*,⁴ you should "use as many levels of headings as you need: no more, and no fewer."

The Tufte- $\text{\LaTeX}$ classes will emit an error if you try to use `\subsubsection` and smaller headings.

IN HIS LATER BOOKS,⁵ Tufte starts each section with a bit of vertical space, a non-indented paragraph, and sets the first few words of the sentence in SMALL CAPS. To accomplish this using this style, use the `\newthought` command:

```
\newthought{In his later books}, Tufte starts...
```

Sidenotes

One of the most prominent and distinctive features of this style is the extensive use of sidenotes. There is a wide margin to provide ample room for sidenotes and small figures. Any `\footnotes` will automatically be converted to sidenotes.⁶ If you'd like to place ancillary information in the margin without the sidenote mark (the superscript number), you can use the `\marginnote` command.

⁴ Robert Bringhurst. *The Elements of Typography*. Hartley & Marks, 3.1 edition, 2005. ISBN 0-88179-205-5

⁵ Edward R. Tufte. *Beautiful Evidence*. Graphics Press, LLC, first edition, May 2006. ISBN 0-9613921-7-7

⁶ This is a sidenote that was entered using the `\footnote` command.

This is a margin note. Notice that there isn't a number preceding the note, and there is no number in the main text where this note was written.

The specification of the `\sidenote` command is:

```
\sidenote[⟨number⟩][⟨offset⟩]{Sidenote text.}
```

Both the `⟨number⟩` and `⟨offset⟩` arguments are optional. If you provide a `⟨number⟩` argument, then that number will be used as the sidenote number. It will change of the number of the current sidenote only and will not affect the numbering sequence of subsequent sidenotes.

Sometimes a sidenote may run over the top of other text or graphics in the margin space. If this happens, you can adjust the vertical position of the sidenote by providing a dimension in the `⟨offset⟩` argument. Some examples of valid dimensions are:

```
1.0in    2.54cm    254mm    6\baselineskip
```

If the dimension is positive it will push the sidenote down the page; if the dimension is negative, it will move the sidenote up the page.

While both the `⟨number⟩` and `⟨offset⟩` arguments are optional, they must be provided in order. To adjust the vertical position of the sidenote while leaving the sidenote number alone, use the following syntax:

```
\sidenote[][⟨offset⟩]{Sidenote text.}
```

The empty brackets tell the `\sidenote` command to use the default sidenote number.

If you *only* want to change the sidenote number, however, you may completely omit the `⟨offset⟩` argument:

```
\sidenote[⟨number⟩]{Sidenote text.}
```

The `\marginnote` command has a similar *offset* argument:

```
\marginnote[⟨offset⟩]{Margin note text.}
```

References

References are placed alongside their citations as sidenotes, as well. This can be accomplished using the normal `\cite` command.⁷

The complete list of references may also be printed automatically by using the `\bibliography` command. (See the end of this document for an example.) If you do not want to print a bibliography at the end of your document, use the `\nobibliography` command in its place.

To enter multiple citations at one location,⁸ you can provide a list of keys separated by commas and the same optional vertical offset argument: `\cite[⟨offset⟩]{Tufte2006,Tufte1990}`.

```
\cite[⟨offset⟩]{bibkey1,bibkey2,...}
```

⁷ The first paragraph of this document includes a citation.

⁸ Edward R. Tufte. *Beautiful Evidence*. Graphics Press, LLC, first edition, May 2006. ISBN 0-9613921-7-7; and Edward R. Tufte. *Envisioning Information*. Graphics Press, Cheshire, Connecticut, 1990. ISBN 0-9613921-1-8

Figures and Tables

Images and graphics play an integral role in Tufte’s work. In addition to the standard `figure` and `tabular` environments, this style provides special figure and table environments for full-width floats.

Full page-width figures and tables may be placed in `figure*` or `table*` environments. To place figures or tables in the margin, use the `marginfigure` or `marginfigure` environments as follows (see figure 1):

```
\begin{marginfigure}
  \includegraphics{helix}
  \caption{This is a margin figure.}
  \label{fig:marginfig}
\end{marginfigure}
```

The `marginfigure` and `marginfigure` environments accept an optional parameter `<offset>` that adjusts the vertical position of the figure or table. See the “[Sidenotes](#)” section above for examples. The specifications are:

```
\begin{marginfigure}[<offset>]
  ...
\end{marginfigure}

\begin{marginfigure}[<offset>]
  ...
\end{marginfigure}
```

`Figure` is an example of the `figure*` environment and `figure` is an example of the normal `figure` environment.

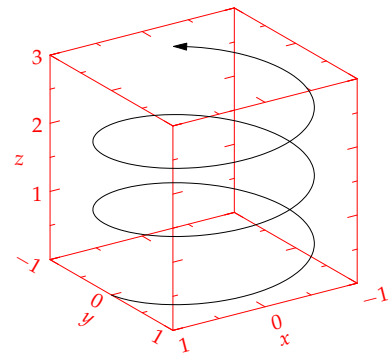


Figure 1: This is a margin figure. The helix is defined by $x = \cos(2\pi z)$, $y = \sin(2\pi z)$, and $z = [0, 2.7]$. The figure was drawn using Asymptote (<http://asymptote.sf.net/>).

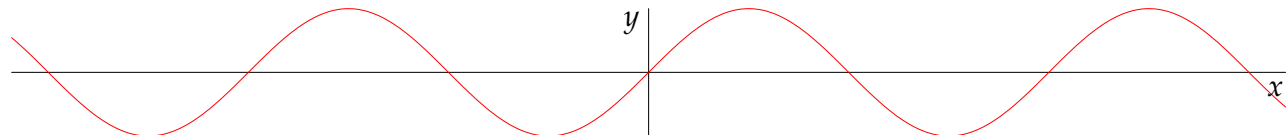


Figure 2: This graph shows $y = \sin x$ from about $x = [-10, 10]$. Notice that this figure takes up the full page width.

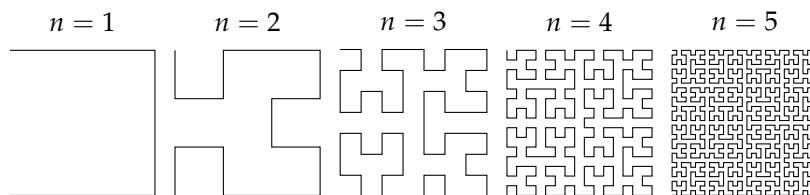


Figure 3: Hilbert curves of various degrees n . Notice that this figure only takes up the main textblock width.

As with `sidenotes` and `marginnotes`, a caption may sometimes require vertical adjustment. The `\caption` command now takes a

second optional argument that enables you to do this by providing a dimension $\langle offset \rangle$. You may specify the caption in any one of the following forms:

```
\caption{long caption}
\caption[short caption]{long caption}
\caption[][ $\langle offset \rangle$ ]{long caption}
\caption[short caption][ $\langle offset \rangle$ ]{long caption}
```

A positive $\langle offset \rangle$ will push the caption down the page. The short caption, if provided, is what appears in the list of figures/tables, otherwise the “long” caption appears there. Note that although the arguments $\langle short\ caption \rangle$ and $\langle offset \rangle$ are both optional, they must be provided in order. Thus, to specify an $\langle offset \rangle$ without specifying a $\langle short\ caption \rangle$, you must include the first set of empty brackets [], which tell `\caption` to use the default “long” caption. As an example, the caption to figure above was given in the form

```
\caption[Hilbert curves...][6pt]{Hilbert curves...}
```

Table shows table created with the `booktabs` package. Notice the lack of vertical rules—they serve only to clutter the table’s data.

Margin	Length
Paper width	8 ¹ / ₂ inches
Paper height	11 inches
Textblock width	6 ¹ / ₂ inches
Textblock/sidenote gutter	3/ ₈ inches
Sidenote width	2 inches

Table 4: Here are the dimensions of the various margins used in the Tufte-handout class.

OCCASIONALLY $\LaTeX$ will generate an error message:

```
Error: Too many unprocessed floats
```

$\LaTeX$ tries to place floats in the best position on the page. Until it’s finished composing the page, however, it won’t know where those positions are. If you have a lot of floats on a page (including sidenotes, margin notes, figures, tables, etc.), $\LaTeX$ may run out of “slots” to keep track of them and will generate the above error.

$\LaTeX$ initially allocates 18 slots for storing floats. To work around this limitation, the Tufte- $\LaTeX$ document classes provide a `\morefloats` command that will reserve more slots.

The first time `\morefloats` is called, it allocates an additional 34 slots. The second time `\morefloats` is called, it allocates another 26 slots.

The `\morefloats` command may only be used two times. Calling it a third time will generate an error message. (This is because we can’t safely allocate many more floats or $\LaTeX$ will run out of memory.)

If, after using the `\morefloats` command twice, you continue to get the Too many unprocessed floats error, there are a couple things you can do.

The `\FloatBarrier` command will immediately process all the floats before typesetting more material. Since `\FloatBarrier` will start a new paragraph, you should place this command at the beginning or end of a paragraph.

The `\clearpage` command will also process the floats before continuing, but instead of starting a new paragraph, it will start a new page.

You can also try moving your floats around a bit: move a figure or table to the next page or reduce the number of sidenotes. (Each sidenote actually uses *two* slots.)

After the floats have placed, $\text{\LaTeX}$ will mark those slots as unused so they are available for the next page to be composed.

Captions

You may notice that the captions are sometimes misaligned. Due to the way $\text{\LaTeX}$'s float mechanism works, we can't know for sure where it decided to put a float. Therefore, the Tufte- $\text{\LaTeX}$ document classes provide commands to override the caption position.

Vertical alignment To override the vertical alignment, use the `\setfloatalignment` command inside the float environment. For example:

```
\begin{figure}[btp]
  \includegraphics{sinewave}
  \caption{This is an example of a sine wave.}
  \label{fig:sinewave}
  \setfloatalignment{b}% forces caption to be bottom-aligned
\end{figure}
```

The syntax of the `\setfloatalignment` command is:

```
\setfloatalignment{<pos>}
```

where `<pos>` can be either `b` for bottom-aligned captions, or `t` for top-aligned captions.

Horizontal alignment To override the horizontal alignment, use either the `\forceversofloat` or the `\forcerectofloat` command inside of the float environment. For example:

```
\begin{figure}[btp]
  \includegraphics{sinewave}
  \caption{This is an example of a sine wave.}
  \label{fig:sinewave}
```

```
\forceversofloat% forces caption to be set to the left of the float
\end{figure}
```

The `\forceversofloat` command causes the algorithm to assume the float has been placed on a verso page—that is, a page on the left side of a two-page spread. Conversely, the `\forcerectofloat` command causes the algorithm to assume the float has been placed on a recto page—that is, a page on the right side of a two-page spread.

Full-width text blocks

In addition to the new float types, there is a `fullwidth` environment that stretches across the main text block and the sidenotes area.

```
\begin{fullwidth}
Lorem ipsum dolor sit amet...
\end{fullwidth}
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Typography

Typefaces

If the Palatino, Helvetica, and Bera Mono typefaces are installed, this style will use them automatically. Otherwise, we'll fall back on the Computer Modern typefaces.

Letterspacing

This document class includes two new commands and some improvements on existing commands for letterspacing.

When setting strings of ALL CAPS or SMALL CAPS, the letterspacing—that is, the spacing between the letters—should be increased slightly.⁹ The `\allcaps` command has proper letterspacing for strings of FULL CAPITAL LETTERS, and the `\smallcaps` command has letterspacing for SMALL CAPITAL LETTERS. These commands will also automatically convert the case of the text to upper- or lowercase, respectively.

The `\textsc` command has also been redefined to include letterspacing. The case of the `\textsc` argument is left as is, however.

⁹ Robert Bringhurst. *The Elements of Typography*. Hartley & Marks, 3.1 edition, 2005. ISBN 0-88179-205-5

This allows one to use both uppercase and lowercase letters: THE INITIAL LETTERS OF THE WORDS IN THIS SENTENCE ARE CAPITALIZED.

Document Class Options

The `tufte-book` class is based on the `LATEX` book document class. Therefore, you can pass any of the typical book options. There are a few options that are specific to the `tufte-book` document class, however.

The `a4paper` option will set the paper size to A4 instead of the default US letter size.

The `sfsidenotes` option will set the sidenotes and title block in a sans serif typeface instead of the default roman.

The `twoside` option will modify the running heads so that the page number is printed on the outside edge (as opposed to always printing the page number on the right-side edge in `oneside` mode).

The `symmetric` option typesets the sidenotes on the outside edge of the page. This is how books are traditionally printed, but is contrary to Tufte's book design which sets the sidenotes on the right side of the page. This option implicitly sets the `twoside` option.

The `justified` option sets all the text fully justified (flush left and right). The default is to set the text ragged right. The body text of Tufte's books are set ragged right. This prevents needless hyphenation and makes it easier to read the text in the slightly narrower column.

The `bidi` option loads the `bidi` package which is used with `XYLATEX` to typeset bi-directional text. Since the `bidi` package needs to be loaded before the sidenotes and cite commands are defined, it can't be loaded in the document preamble.

The `debug` option causes the Tufte-`LATEX` classes to output debug information to the log file which is useful in troubleshooting bugs. It will also cause the graphics to be replaced by outlines.

The `nofonts` option prevents the Tufte-`LATEX` classes from automatically loading the Palatino and Helvetica typefaces. You should use this option if you wish to load your own fonts. If you're using `XYLATEX`, this option is implied (*i.e.*, the Palatino and Helvetica fonts aren't loaded if you use `XYLATEX`).

The `nols` option inhibits the letterspacing code. The Tufte-`LATEX` classes try to load the appropriate letterspacing package (either `pdfTEX`'s `letterspace` package or the `soul` package). If you're using `XYLATEX` with `fontenc`, however, you should configure your own letterspacing.

The `notitlepage` option causes `\maketitle` to generate a title

block instead of a title page. The book class defaults to a title page and the handout class defaults to the title block. There is an analogous **titlepage** option that forces `\maketitle` to generate a full title page instead of the title block.

The **notoc** option suppresses Tufte- $\text{\LaTeX}$'s custom table of contents (toc) design. The current toc design only shows unnumbered chapter titles; it doesn't show sections or subsections. The notoc option will revert to $\text{\LaTeX}$'s toc design.

The **nohyper** option prevents the hyperref package from being loaded. The default is to load the hyperref package and use the `\title` and `\author` contents as metadata for the generated pdf.

Customizing Tufte- $\text{\LaTeX}$

The Tufte- $\text{\LaTeX}$ document classes are designed to closely emulate Tufte's book design by default. However, each document is different and you may encounter situations where the default settings are insufficient. This chapter explores many of the ways you can adjust the Tufte- $\text{\LaTeX}$ document classes to better fit your needs.

File Hooks

If you create many documents using the Tufte- $\text{\LaTeX}$ classes, it's easier to store your customizations in a separate file instead of copying them into the preamble of each document. The Tufte- $\text{\LaTeX}$ classes provide three file hooks: `tufte-common-local.tex`, `tufte-book-local.tex`, and `tufte-handout-local.tex`.

tufte-common-local.tex If this file exists, it will be loaded by all of the Tufte- $\text{\LaTeX}$ document classes just prior to any document-class-specific code. If your customizations or code should be included in both the book and handout classes, use this file hook.

tufte-book-local.tex If this file exists, it will be loaded after all of the common and book-specific code has been read. If your customizations apply only to the book class, use this file hook.

tufte-common-handout.tex If this file exists, it will be loaded after all of the common and handout-specific code has been read. If your customizations apply only to the handout class, use this file hook.

Numbered Section Headings

While Tufte dispenses with numbered headings in his books, if you require them, they can be enabled by changing the value of the `secnumdepth` counter. From the table below, select the heading level at which numbering should stop and set the `secnumdepth` counter to that value. For example, if you want parts and chapters

numbered, but don't want numbering for sections or subsections, use the command:

```
\setcounter{secnumdepth}{0}
```

The default `secnumdepth` for the Tufte- $\text{\LaTeX}$ document classes is `-1`.

Heading level	Value
Part (in tufte-book)	-1
Part (in tufte-handout)	0
Chapter (only in tufte-book)	0
Section	1
Subsection	2
Subsubsection	3
Paragraph	4
Subparagraph	5

Table 5: Heading levels used with the `secnumdepth` counter.

Changing the Paper Size

The Tufte- $\text{\LaTeX}$ classes currently only provide three paper sizes: A4, B5, and US letter. To specify a different paper size (and/or margins), use the `\geometrysetup` command in the preamble of your document (or one of the file hooks). The full documentation of the `\geometrysetup` command may be found in the `geometry` package documentation.¹⁰

¹⁰ Hideo Umeki. The geometry package. <http://ctan.org/pkg/geometry>, December 2008

Customizing Marginal Material

Marginal material includes sidenotes, citations, margin notes, and captions. Normally, the justification of the marginal material follows the justification of the body text. If you specify the `justified` document class option, all of the margin material will be fully justified as well. If you don't specify the `justified` option, then the marginal material will be set ragged right.

You can set the justification of the marginal material separately from the body text using the following document class options: `sidenote`, `marginnote`, `caption`, `citation`, and `marginals`. Each option refers to its obviously corresponding marginal material type. The `marginals` option simultaneously sets the justification on all four marginal material types.

Each of the document class options takes one of five justification types:

justified Fully justifies the text (sets it flush left and right).

raggedleft Sets the text ragged left, regardless of which page it falls on.

raggedright Sets the text ragged right, regardless of which page it falls on.

raggedouter Sets the text ragged left if it falls on the left-hand (verso) page of the spread and otherwise sets it ragged right. This is useful in conjunction with the *symmetric* document class option.

auto If the justified document class option was specified, then set the text fully justified; otherwise the text is set ragged right. This is the default justification option if one is not explicitly specified.

For example,

```
\documentclass[symmetric,justified,marginals=raggedouter]{tufte-book}
```

will set the body text of the document to be fully justified and all of the margin material (sidenotes, margin notes, captions, and citations) to be flush against the body text with ragged outer edges.

THE FONT AND STYLE of the marginal material may also be modified using the following commands:

```
\setsidenotefont{<font commands>}
\setcaptionfont{<font commands>}
\setmarginnotefont{<font commands>}
\setcitationfont{<font commands>}
```

The `\setsidenotefont` sets the font and style for sidenotes, the `\setcaptionfont` for captions, the `\setmarginnotefont` for margin notes, and the `\setcitationfont` for citations. The ** can contain font size changes (e.g., `\footnotesize`, `\Huge`, etc.), font style changes (e.g., `\sffamily`, `\ttfamily`, `\itshape`, etc.), color changes (e.g., `\color{blue}`), and many other adjustments.

If, for example, you wanted the captions to be set in italic sans serif, you could use:

```
\setcaptionfont{\itshape\sffamily}
```


Compatibility Issues

When switching an existing document from one document class to a Tufte- $\text{\LaTeX}$ document class, a few changes to the document may have to be made.

Converting from article to tufte-handout

The following article class options are unsupported: 10pt, 11pt, 12pt, a5paper, b5paper, executivepaper, legalpaper, landscape, onecolumn, and twocolumn.

The following headings are not supported: $\backslash$ subsubsection and $\backslash$ subparagraph.

Converting from book to tufte-book

The following report class options are unsupported: 10pt, 11pt, 12pt, a5paper, b5paper, executivepaper, legalpaper, landscape, onecolumn, and twocolumn.

The following headings are not supported: $\backslash$ subsubsection and $\backslash$ subparagraph.

Troubleshooting and Support

Tufte- $\text{\LaTeX}$ Website

The website for the Tufte- $\text{\LaTeX}$ packages is located at <http://code.google.com/p/tufte-latex/>. On our website, you'll find links to our svn repository, mailing lists, bug tracker, and documentation.

Tufte- $\text{\LaTeX}$ Mailing Lists

There are two mailing lists for the Tufte- $\text{\LaTeX}$ project:

Discussion list The tufte-latex discussion list is for asking questions, getting assistance with problems, and help with troubleshooting. Release announcements are also posted to this list. You can subscribe to the tufte-latex discussion list at <http://groups.google.com/group/tufte-latex>.

Commits list The tufte-latex-commits list is a read-only mailing list. A message is sent to the list any time the Tufte- $\text{\LaTeX}$ code has been updated. If you'd like to keep up with the latest code developments, you may subscribe to this list. You can subscribe to the tufte-latex-commits mailing list at <http://groups.google.com/group/tufte-latex-commits>.

Getting Help

If you've encountered a problem with one of the Tufte- $\text{\LaTeX}$ document classes, have a question, or would like to report a bug, please send an email to our mailing list or visit our website.

To help us troubleshoot the problem more quickly, please try to compile your document using the debug class option and send the generated .log file to the mailing list with a brief description of the problem.

Errors, Warnings, and Informational Messages

The following is a list of all of the errors, warnings, and other messages generated by the Tufte- $\text{\LaTeX}$ classes and a brief description of their meanings.

Error: `\subparagraph` is undefined by this class.

The `\subparagraph` command is not defined in the Tufte- $\text{\LaTeX}$ document classes. If you'd like to use the `\subparagraph` command, you'll need to redefine it yourself. See the "Headings" section on page 27 for a description of the heading styles available in the Tufte- $\text{\LaTeX}$ document classes.

Error: `\subsubsection` is undefined by this class.

The `\subsubsection` command is not defined in the Tufte- $\text{\LaTeX}$ document classes. If you'd like to use the `\subsubsection` command, you'll need to redefine it yourself. See the "Headings" section on page 27 for a description of the heading styles available in the Tufte- $\text{\LaTeX}$ document classes.

Error: You may only call `\morefloats` twice. See the Tufte- $\text{\LaTeX}$ documentation for other workarounds.

$\text{\LaTeX}$ allocates 18 slots for storing floats. The first time `\morefloats` is called, it allocates an additional 34 slots. The second time `\morefloats` is called, it allocates another 26 slots.

The `\morefloats` command may only be called two times. Calling it a third time will generate this error message. See page 30 for more information.

Warning: Option '`\langle class option \rangle`' is not supported -- ignoring option.

This warning appears when you've tried to use `\langle class option \rangle` with a Tufte- $\text{\LaTeX}$ document class, but `\langle class option \rangle` isn't supported by the Tufte- $\text{\LaTeX}$ document class. In this situation, `\langle class option \rangle` is ignored.

Info: The 'symmetric' option implies 'twoside'

You specified the symmetric document class option. This option automatically forces the twoside option as well. See page 33 for more information on the symmetric class option.

Package Dependencies

The following is a list of packages that the Tufte- $\text{\LaTeX}$ document classes rely upon. Packages marked with an asterisk are optional.

- xifthen
- ifpdf*
- ifxetex*
- hyperref
- geometry
- ragged2e
- chngpage *or* changepage
- paralist
- textcase
- soul*
- letterspace*
- setspace
- natbib *and* bibentry
- optparams
- placeins
- mathpazo*
- helvet*
- fontenc
- beramono*
- fancyhdr
- xcolor
- textcomp
- titlesec
- titletoc

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